



Fig (8.13)(a)
Biomass energy (Wood)



Fig (8.13)(b)
Biomass energy
(Organic materials)



Fig (8.13)(c)
Biomass energy (Garbage)



Fig (8.14)
Tidal energy

- ◆ It is being used by wind mills to grind grain.
- ◆ It is used to turn wind turbines to produce electricity.

ii. Biomass Energy

Biomass is the organic material that comes from plants and animals. Biomass consists of stored energy from Sun, garbage, wastes, sugarcane etc. Solid biomass, such as wood, organic material and garbage, can be burned directly to produce heat; Fig (8.13) (a, b, c). Biomass can also be converted into gas called biogas and into liquid biofuels such as ethanol and biodiesel.

viii. Tidal Energy

It is a form of hydro power that converts the energy obtained from tides into useful form of power; mainly electricity as the Earth uses the gravitational forces of both the moon and the sun every day to move vast quantities of water around the oceans and seas producing tides and in this way energy is produced called tidal energy; (Fig 8.14).

Self Assessment Questions:

Q7: What is biomass?

Q8: Write down the name of fossil fuel.

Q9: Which type of energy is stored deep in the Earth?

8.4 RENEWABLE AND NON-RENEWABLE ENERGY SOURCE

Renewable Energy Source

The renewable sources can be consumed and used again and again. Solar energy, wind energy, tidal energy and geothermal energy are renewable sources.



Since very earlier age, people have tried to consume renewable sources of energy for their survival. Such as wind and water for milling grain and solar for lighting.

Non-Renewable Sources

Non-renewable resources are limited and will finish once used. Coal, petroleum and natural gases are nonrenewable sources. About 150 years ago scientists invented new technology to extract energy from the ancient fossilized remains of plants and animals. These super-rich but limited sources of energy (coal, oil and natural gas) replaced wood, wind and water as the main sources of fuel. They are being used at a faster rate than they can be restored again and, therefore cannot be renewed.

Self Assessment Questions:

Q10: Write down the names of any three renewable energy sources.

Q11: Write down the names of any three non-renewable energy sources.

Q12: What is the difference between renewable and non-renewable energy sources?

8.5 EFFICIENCY

Every machine needs some energy to perform work. Whatever energy given to a machine is called input and the work done by the machine is called output.

For example: We give electric energy as input to the electric motor in washing machines and in drilling machines.



Do You Know!

- ◆ Wind energy is clean fuel source.
- ◆ It does not pollute the air.
- ◆ Wind turbines does not produce atmospheric emission that causes greenhouse gasses.



Do You Know!

The costal belt of Pakistan is about 1045 km long with best resources for utilizing and producing tidal energy.



A system in which some energy ' E_1 ' is supplied to it as 'input' and the system returns back some energy ' E_2 ' as output has some efficiency. This efficiency is defined as

The ratio of output to the input is called Efficiency.

Efficiency is denoted by " η " called eta. As it is the ratio of two energies therefore it has no unit. No machine is 100% efficient because some energy is always wasted in the form of heat, sound or light etc.



Do You Know!

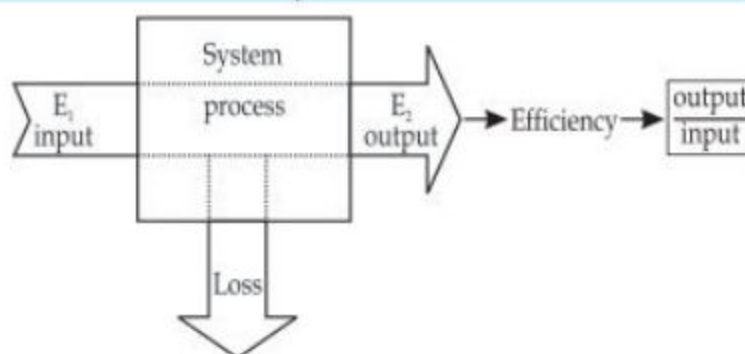
We get

- ◆ Chemical energy from fuel, gas and battery.
- ◆ Thermal energy from hot object.
- ◆ Nuclear energy from nuclear Fission and fusion.
- ◆ Electrical energy from movement of charges.
- ◆ Mechanical energy from walking, running.
- ◆ Sound energy from sound waves.

$$\text{Efficiency} = \frac{\text{Energy as output } (E_2)}{\text{Energy as input } (E_1)}$$

$$\therefore \eta = \frac{E_2}{E_1} \times 100$$

$$\text{Efficiency} = \frac{\text{output}}{\text{input}} \times 100 \dots\dots\dots(8.5)$$



8.6 POWER

When you run up and cover distance in 5 seconds or take slow walk up the same distance in 20 seconds. You are doing the same amount of work, However, you are doing it at different rate. When you run up, you are working much faster and you have a higher power then when you walk up.

This quantity that tells us the rate of doing work. Thus, power is defined as:



The rate of doing work, or
The amount of energy transferred per unit time.
Mathematically,

$$\text{Power} = P = \frac{\text{work done}}{\text{time taken}}$$

$$\therefore P = \frac{W}{t} \dots\dots\dots(8.6)$$

Since work and time are scalar quantities. Therefore, power is also a scalar quantity.

Unit of Power

In SI system unit of power is $\frac{\text{Joule}}{\text{sec}} = \text{Watt}$

Thus SI unit of power is watt which is defined as:

The power of a body is said to be one watt if it does work at the rate of one Joule per second.

Worked Example 5

Calculate the power of a machine. If the machine performs 900 joules of work in 30 minutes.

Solution

Step 1: Write down known quantities and quantities to be found.

$$W = 900\text{J}$$

$$t = 30\text{min} = 30 \times 60\text{s} = 1800\text{s}$$

$$P = ??$$

Step 2: Write down formula and rearrange if necessary

$$P = \frac{W}{t}$$

Step 3: Put values in formula and calculate

$$P = \frac{900\text{J}}{1800\text{s}}$$

$$P = 0.5\text{ W}$$

Hence, power of the machine is 0.5 Watt.



Do You Know!

1kg of 4% enriched fuel grade uranium releases energy equivalent to the combustion of nearly 100 tons of high grade coal or 60 tons of oil.



Do You Know!

Larger units of power are

Kilo watt (kW),
Mega watt (MW),
Horse Power (hp) etc.
 $1\text{kW} = 1000\text{W} = 10^3 \text{ watt}$
 $1\text{MW} = 1000000\text{W} = 10^6 \text{ Watt}$
 $1 \text{ hp} = 746 \text{ Watt}$



Do You Know!

$$P = \frac{W}{t}$$

$$P = \frac{F \cdot d}{t}$$

$$P = F \cdot \frac{d}{t} \because \frac{d}{t} = V$$

$$P = F \cdot V$$



Self Assessment Questions:

Q13: A man pushes a car 18m with a force of 2N in 4 second. Calculate the power of the man.

Q14: Why power is a scalar quantity?

Q15: Name the physical quantity which gives the rate of doing work.

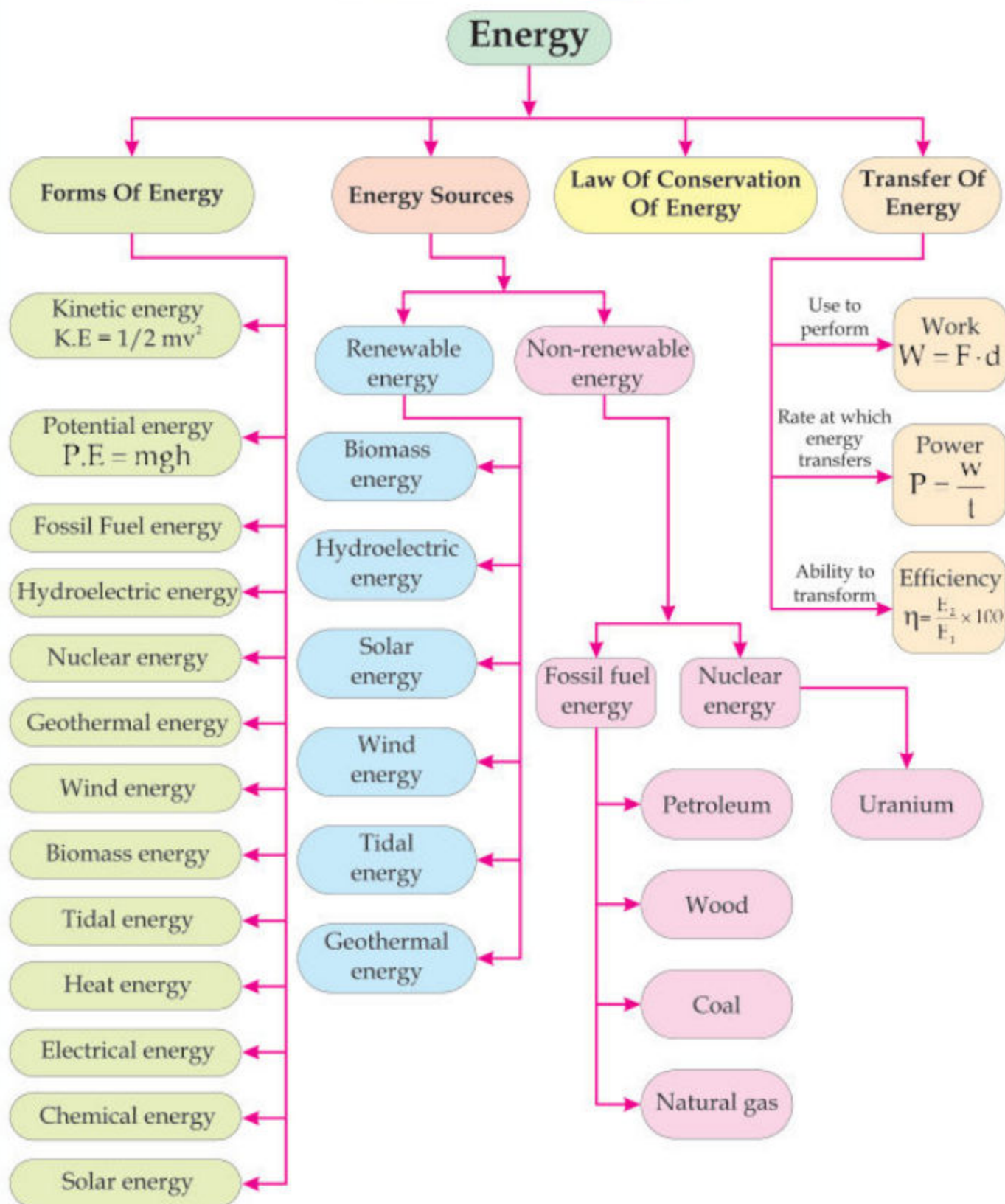


SUMMARY

- ◆ Work is the product of force and the distance ($W = F \cdot S$)
- ◆ The ability to do work is called energy. SI unit of energy is Joule (J).
- ◆ Energy possessed by an object due to its motion is called as Kinetic Energy ($K \cdot E = \frac{1}{2}mv^2$).
- ◆ Energy due to position of an object is called Potential Energy ($P \cdot E = mgh$).
- ◆ Energy exists in many different forms such as nuclear energy, heat energy, electrical energy, chemical energy, light energy, etc.
- ◆ Solar energy, wind energy, tidal energy, geothermal energy, biomass energy and hydroelectric energy are the examples of renewable sources of energy.
- ◆ Wood, coal, petroleum, natural gas and Uranium are examples of nonrenewable sources of energy.
- ◆ The ratio of output to the input is called efficiency.
- ◆ The work done in unit time is called power. SI unit of work is (Watt).



CONCEPT MAP





End of Unit Questions

Section (A) Multiple Choice Questions (MCQs)

1. If force of 6N displaces an object 2m in the direction of force, then work done will be _____.
a) 0 Joule b) 12 Joule
c) 3 Joule d) 12 N.
2. If a body of mass 1 kg is moving with velocity of 1m/sec then K.E of the body will be _____.
a) 10 Joules b) 0.5 Joules
c) 2 Joules d) 1 Joule
3. If a machine performs 20J of work in 10sec then its power is _____.
a) 200 watt b) 20 watt
c. 2watt d) 0.2 watt
4. A body of mass 1kg is lifted through a height of 1m . The energy possessed in the body will be _____. (consider $g = 10\text{ms}^{-2}$)
a) 1J b) 10Joule
c) 100Joule d) 1000Joule
5. The energy released during fission or fusion reaction is called _____.
a) Solar energy b) Geothermal energy
c) Tidal energy d) Nuclear energy
6. Which is the renewable source of energy
a) Solar and wind b) Coal
c) Natural gas d) Petroleum
7. The ratio of output to input is called
a) Energy b) Work
c) Power d) Efficiency



8. Work done per unit time is called _____.
 a) Efficiency b) Energy
 c) Power d) Force
9. Coal, gas and oil are all examples of _____.
 a) Tidal energy b) Nuclear energy
 c) Fossil fuel energy d) Biomass energy
10. _____ is not a renewable source of energy.
 a) Solar energy b) Coal
 c) Wind energy d) Geothermal energy

Section (B) Structured Questions

Work

- Define work.
 - Derive the equation; ($\text{Work} = Fd \cos\theta$).
- How much work is needed to move horizontally a body 20m by a force of 30N, the angle between the body and the horizontal surface is 60° ?
- How much work is done, if a crate is moved at a distance of 50m, when a force of 30N is applied along the surface.
- What is the work done by Usman? If a bar of weight 100 N is brought by him from (A) to (B), then brought back to (A).

Energy Forms

- Define Kinetic energy
 - Derive the equation ($K.E = \frac{1}{2}mv^2$).
- What will be the Kinetic energy of a boy of mass 50kg driving a bike with velocity of 2ms^{-1} ?
- Define Potential Energy.
 - Derive the equation ($P.E = mgh$).



8. a) If LED screen of mass 10kg is lifted up and kept it on a cupboard of height 2m. Calculate the potential energy stored in the LED screen.
b) Calculate the potential energy of 3kg water raised to the tank at the roof of a home 4m high. (assume $g=10\text{ms}^{-2}$)

Conversion of Energy

9. a) Why fossil fuel energy is called non-renewable source?
b) Define solar energy and its importance in Pakistan.
10. Write notes on Tidal energy and Geothermal energy.
11. a) What is wind energy?
b) Write any three applications of wind energy.
12. a) Write the name of any one radioactive element which is used as source of nuclear energy.
b) Write the names of any one device that can convert solar energy into heat energy.
c) Write the names of any two devices that can convert solar energy into electrical energy.

Renewable and Non-renewable Energy Sources

13. Write a note on renewable energy sources.
14. Write a note on non-renewable energy sources.
15. What is the difference between renewable of non-renewable energy sources?
16. Make a table of renewable and non-renewable energy sources from the following:
Uranium, Solar, Coal, Wind, Natural gas, Tidal, Biomass, Hydroelectricity.



Efficiency

17. Calculate the efficiency of a machine which consumes 200 J of energy and performs 50 J of work.
18. Write a note on efficiency.
19. If the efficiency of a machine is 70% and its output is 100 J then calculate its input.
20. Which machine is more efficient, machine "A" which has an output of 200 J after consuming 400 J of energy or machine "B" which has an output of 300 J after consuming 450 J of energy?

Power

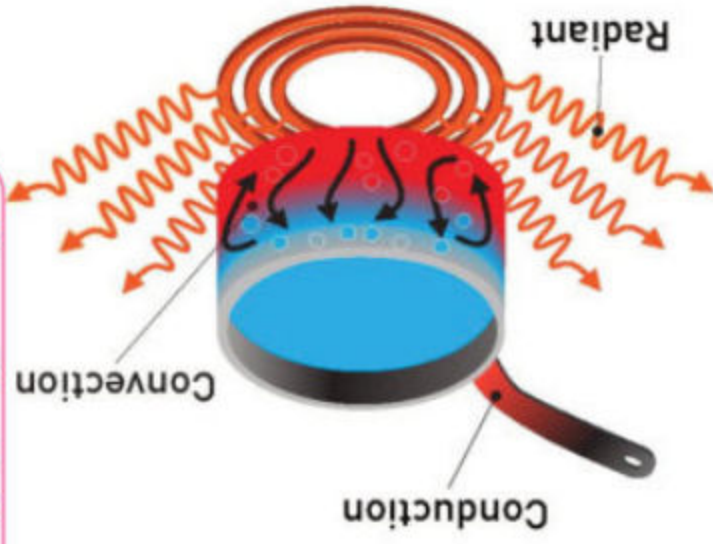
21. a) Define power.
b) The energy of 600 J dissipated by a bulb in 50 minutes. Find the power of the bulb.
22. a) Convert 20 watt into horse power.
b) Calculate the power of a machine, if it does 40 Joules of work in 10 sec.
23. a) Define Watt.
b) A student of weight 400 N takes 5 sec to climb up an obstacle of height 2 m. Calculate the power consumed.
24. a) Write down the names of any two larger units of power.
b) If a machine consumes 250 J of energy per hour then what will be its power?

THERMAL PROPERTIES OF MATTER

Unit - 9



- Students Learning Outcomes (SLOs)
- After learning this unit students should be able to:
- Differentiate between heat and temperature.
 - Define the terms heat capacity and specific heat capacity with SI unit.
 - Describe one everyday effect due to relatively large specific heat of water.
 - Describe heat of fusion and heat of vaporization (as energy transfer without a change of temperature for change of state).
 - Describe experiments to determine heat of fusion and heat of vaporization of ice and water respectively by sketching temperature-time graph on heating ice.
 - Explain the process of evaporation and the difference between boiling and evaporation.
 - Explain that evaporation causes cooling.
 - List the factors which influence surface evaporation.
 - Define thermal expansion.
 - Describe qualitatively the thermal expansion of solids (linear and volumetric expansion).
 - List and explain some of the everyday applications and consequences of thermal expansion.
 - Explain the thermal expansion of liquids (real and apparent expansion).



The properties of matter which change with the change in temperature are called "Thermal properties". Materials change their states due to change in their temperature, cause by the addition or removal of heat. Like for example, water changes from liquid to gas 'steam' due to addition of heat or to solid 'ice' due to removal of heat. The process of evaporation, boiling, freezing, thermal expansion (linear and volumetric) are all happens due to the change in temperature of the substance. Thus changes in temperature caused by the addition or removal of heat play very important role in our daily life.



The objective of this unit is to create critical logical thinking among the students, so that they can observe and analyze the physical quantities and changes taking place in their surroundings.

- How does water in our surrounding helps us to maintain the temperature of environment?
- Why different liquids heat up in different manner at same time and same temperature?
- What is the importance of specific heat in everyday life?
- How evaporation takes place and its affects on the surroundings?
- What happen when temperature of solids and liquids increases?
- How does physical state of matter changes from one state to other?
- Why wet clothes take more time to dry out in cloudy day or in winter?
- What is the role of thermal expansion in our daily life?

9.1 HEAT AND TEMPERATURE

Heat



Fig (9.1) (a)
Heat

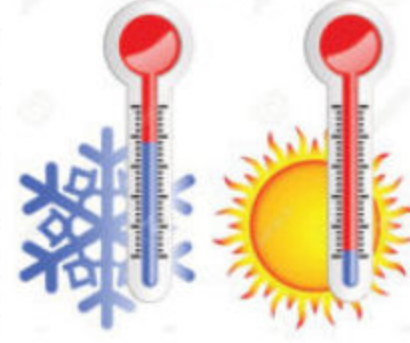


Fig (9.1) (b)
Temperature

Mostly we think that heat and temperature are the same, however, this is not true, they are related with each other, but are different concepts; Fig (9.1)(a, b). Heat is a form of energy. Where as temperature is degree of hotness or coldness. For Example: A hot cup of tea is placed on table, after some time the tea in the cup becomes cold because surrounding temperature is lower than that of the hot tea. Hence heat flows from hot cup to the surrounding. Therefore heat and temperature can be stated as:



Heat	It is the form of energy which transfers from hot body to cold body as a result of difference of temperature between them. As heat is form of energy. Therefore its SI unit is (joule). Its other unit is (calorie).
Temperature	It is a degree of hotness and coldness of a body. It determines the direction of flow of heat from one body to the other body. SI unit of temperature is (Kelvin). Its other units are (Celsius) and (Fahrenheit).

Thermometer

Thermometer is a device, used to measure temperature.
For example;
A clinical thermometer is used to measure the temperature of human body (Fig 9.2).



Fig (9.2)
Clinical thermometer

Thermometers have different scales to measure temperature.

- There are three scales of temperature (Fig 9.3).
- 1. Celsius scale (Mostly used for environmental measurements)
 - 2. Fahrenheit scale (Mostly used for clinical measurements)
 - 3. Kelvin scale (Mostly used for industrial measurements)